Hongbo Ma

Xinya College & Department of Computer Science and Technology, Tsinghua University
+86-133-5868-9152 | mahb22@mails.tsinghua.edu.cn | ma-hongbo.github.io

Education

Tsinghua University Sep 2022 – Present
B.S. in Computer Science and Technology, GPA: 3.8/4.0 *Beijing, China*
Focus: machine learning, computer vision, and mathematical foundations

Tsinghua University Sep 2023 – Present
Minor in Economics and Management, GPA: 3.8/4.0 *Beijing, China*
Focus: economics, management, and investment

Research Interests

Building Efficient, Interpretable and Reliable Decision-Making Systems for Cross-Modal Intelligence.

- Real2Sim–Sim2Sim–Sim2Real research loop driven by **world models**, **generative models** , and **multi-agent system**.
- Based on **LLMs/VLMs**, focus on embodied perception-reasoning-acting, efficient decision-making under real-world constraints.

Publications *(* denotes equal contribution)*

Major

- Bangji Yang*, **Hongbo Ma***, Qinglong Zheng*, Jiajun Fan, JIA-HONG HUANG, Ge Liu. **Auditing Generated Skill Libraries for Embodied Language Agents.** ACL ARR 2026 May, Under Review
- Bangji Yang*, Jiajun Fan*, **Hongbo Ma***, Qinglong Zheng, Ge Liu. **From Melee to Consensus: Adaptive Moderated Debate for Inference-Time LLM Reasoning.** ACL ARR 2026 May, Under Review
- Bangji Yang*, **Hongbo Ma***, Ruihan Liu, Jiajun Fan, Xi Zhu, Ge Liu. **Concurrent Constraint Reinforcement.** Neurips 2026, Under Review
- Bangji Yang*, **Hongbo Ma***, Jiajun Fan, Ge Liu. **Batched Contextual Reinforcement. ICML 2026, Poster**
- **Hongbo Ma**, Fei Shen, Hongbin Xu, Xiaoce Wang, Gang Xu, Jinkai Zheng, Liangqiong Qu, Ming Li. **StyleTailor: Towards Personalized Fashion Styling via Hierarchical Negative Feedback. AAAI 2026, Oral**

Secondary

- Kuan Zhang*, Dongchen Liu*, Qiyue Zhao*, Tianyu Xin*, Yue Su*, Haisheng Wang, Han Yin, **Hongbo Ma**, Peize Li, Tianjun Gu, Xiangnan Wu, Xinran Zhang, Yongxuan Li, Zirong Chen, Yiming Li. **Towards Generalist Game Players: A Survey of Foundation Models in the Game Multiverse.** TMLR, Under Review
- Juncheng Yan, Qi Qin, Yinan Liang, **Hongbo Ma**, Le Zhuo, Yuandong Pu, Yi Xin, Wenzhao Zheng, Yu Qiao, Yihao Liu, Jiwen Lu. **Towards High-Fidelity Mobile Video Synthesis with Slimmed Diffusion-Transformer.** ECCV 2026, Under Review
- Keyue Qiu*, Yuxuan Song*, Jie Yu, **Hongbo Ma**, Ziyao Cao, Mingyue Zheng, Hao Zhou, Wei-Ying Ma. **Empower Structure-based Molecule Optimization with Gradient Guidance. ICML 2025, Poster**

- Yanwen Huang*, Bowen Gao*, Yinjun Jia, **Hongbo Ma**, Wei-Ying Ma, Ya-Qin Zhang, Yanyan Lan. **Redefining the Task of Bioactivity Prediction. ICLR 2025, Poster**

Research Experience

Siebel School of Computing and Data Science, UIUC Dec 2025 – Present
Research Intern, advised by Prof. Ge Liu *Champaign, IL, USA*

- To mitigate the computational redundancy in multi-step reasoning, formulated grouped mathematical reasoning as a batched RL problem and proposed Batched Contextual Reinforcement (BCR) to share context across samples, achieving a 2x throughput enhancement by reducing token usage by 50% without compromising inference accuracy.
- Built the reward design and TRL training pipeline for mathematical LLM post-training, iterating on objectives with attention to cost, reasoning efficiency, stability, and credit assignment.

i-Vision Group, Department of Automation, Tsinghua University Sep 2025 – Oct 2025
Research Intern, advised by Prof. Jiwen Lu *Beijing, China*

- Motivated by deployment cost-performance tradeoffs, established a holistic framework for algorithm-system co-design, architecting a slimmed Diffusion Transformer that reconciles high-fidelity generation with mobile-side VRAM constraints.
- Deployed Swift and MLX inference pipelines, using profiling-driven optimization to turn architectural compression into cost-efficient on-device speedups.

Guangming Lab May 2025 – Aug 2025
Research Intern, advised by Prof. Ming Li *Shenzhen, China*

- Driven by the belief that AI should serve daily life, orchestrated a closed-loop alignment framework with hierarchical negative feedback to resolve ambiguity in one-shot fashion recommendations.
- Structured the end-to-end multi-agent workflow for design, recommendation, try-on, and evaluation, turning feedback into a deployable fashion-assistance pipeline; led writing, figures, and rebuttal as sole first author for an AAAI 2026 oral paper.

GenSi Lab, AIR Tsinghua Jul 2024 – Feb 2025
Research Intern, advised by Prof. Hao Zhou *Beijing, China*

- Engineered PyTorch generative pipelines from VAE and diffusion foundations, motivated by the AI-biopharma frontier, and used those priors to reason about stability and controllability in molecular generation.
- Approached structure-based molecule optimization as a joint problem over continuous coordinates and discrete atom types, and developed a Bayesian Flow Networks framework to orchestrate coherent gradient guidance, bridging continuous chemical spaces with discrete molecular topologies.

ATOM Lab, AIR Tsinghua Feb 2024 – Jun 2024
Research Intern, advised by Prof. Yanyan Lan *Beijing, China*

- Investigated why bioactivity models overfit spurious pocket cues, treating reliable evaluation as foundational to AI-biopharma rather than a narrow model issue.
- Implemented preprocessing, training, and evaluation workflows, including statistical analysis and visualization, to support reliable molecule-level assessment for bioactivity prediction in this emerging space.

Selected Project

AI-Enabled Production and Dissemination of Traditional Culture 2024 – 2025
National Undergraduate Innovation and Entrepreneurship Program *Tsinghua University*

- Co-led an AIGC project for traditional-culture video production, motivated by communication effectiveness, and reconceptualized cultural dissemination as an end-to-end generative pipeline, ensuring narrative integrity from conceptual planning to multi-platform release.

Services

- **Neurips 2026**, Reviewer
- **ICML 2026 Workshop AI4Math**, Reviewer

Leadership

- **President, Student Union**, Xinya College, Tsinghua University

Awards

- **Outstanding Scientific and Technological Innovation Scholarship**, Tsinghua University (2024 – 2025)
- **Outstanding Student Leader**, Tsinghua University (2023 – 2024)
- **Outstanding Social Work Scholarship**, Tsinghua University (2022 – 2023, 2023 – 2024, 2024 – 2025)
- **Second Prize Freshman Scholarship**, Tsinghua University (2022)

Skills

- **Programming:** Python, C/C++, Swift, MATLAB, Rust, Java, TypeScript/JavaScript, SystemVerilog
- **ML Stack:** PyTorch, NumPy, MLX, diffusion and flow-based models, PPO/DPO/GRPO pipelines
- **Languages:** Chinese (Native), English (Fluent)